Prayas JEE AIR (2025)

WEEKLY TEST 31

DURATION : 180 Minutes | **DATE : 16/01/2025** | **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25** questions in each part in which first **20** questions are of Objective Type and Last **5** questions are integers type and all **25** questions are **compulsory.**
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number: _____

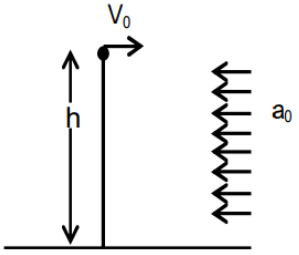
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Candidate's Signature: _____ **Invigilator's Signature** _____

SECTION-I (PHYSICS)

Single Correct Type Questions

1. A particle is projected horizontally from a tower with velocity V_0. Wind is blowing in opposite direction and is providing a constant horizontal acceleration a_0. If particle strikes the ground vertically then height of building is



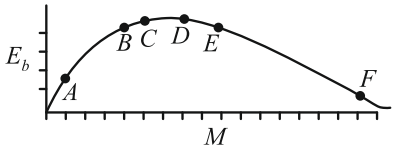
(1) $\frac{V_0^2 g}{a_0^2}$ (2) $\frac{V_0^2 g}{2a_0^2}$

(3) $\frac{2V_0^2 g}{a_0^2}$ (4) $\frac{V_0^2}{g}$

2. The above is a plot of binding energy per nucleon E_b, against the nuclear mass M; A, B, C, D, E, F correspond to different nuclei. Consider four reactions:

(i) $A + B \rightarrow C$ + Energy (E_1)
(ii) $C \rightarrow A + B$ + Energy (E_2)
(iii) $D + E \rightarrow F$ + Energy (E_3)
(iv) $F \rightarrow D + E$ + Energy (E_4)

In which reactions energy released (positive)?



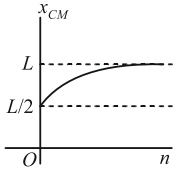
(1) (i) and (iv) (2) (i) and (iii)
(3) (ii) and (iv) (4) (ii) and (iii)

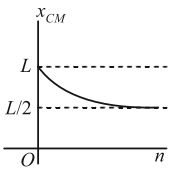
3. A thin uniform rod of length l and mass m is swinging freely about a horizontal axis passing through its end. Its maximum angular speed is ω. Its centre of mass rises to a maximum height of

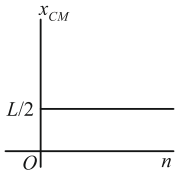
(1) $\frac{1}{3}\frac{l^2\omega^2}{g}$ (2) $\frac{l^3\omega^4}{6g^2}$

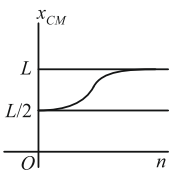
(3) $\frac{1}{2}\frac{l^2\omega^2}{g}$ (4) $\frac{1}{6}\frac{l^2\omega^2}{g}$

4. A thin rod of length 'L ' is lying along the x-axis with its ends at $x = 0$ and $x = L$. Its linear density $\lambda(x)$ (mass/length) varies with x as $\lambda(x) = k\left(\frac{x}{L}\right)^n$, where n can be zero or any positive number. If the position x_{CM} of the centre of mass of the rod is plotted against 'n', which of the following graphs best approximates the dependence of x_{CM} on n?

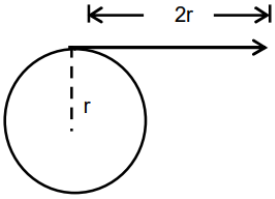
(1) 

(2) 

(3) 

(4) 

5. A uniform bar of mass m and length $2r$ is attached to ring of mass m and radius r. such that it is tangential to the ring and both are in same plane. Moment of inertia of this assembly about an axis passing through centre of ring and perpendicular to plane is



(1) $\frac{10}{3}mr^2$ (2) $\frac{7}{3}mr^2$

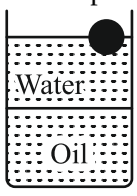
(3) $\frac{11}{3}mr^2$ (4) $\frac{19}{3}mr^2$

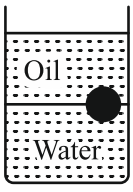
6. A planet of mass m moves along an ellipse around the sun so that its maximum and minimum distances from the sun are equal to R and r respectively. The angular momentum of this planet relative to the centre of the sun (mass M) is

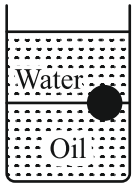
(1) $2\sqrt{\frac{GMmrR}{r+R}}$ (2) $2R\sqrt{\frac{GMmR}{r+R}}$

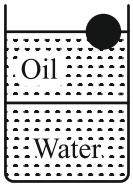
(3) $2m\sqrt{\frac{GMrR}{r+R}}$ (4) $m\sqrt{\frac{2GMrR}{r+R}}$

7. A ball is made of a material of density ρ where $\rho_{oil} < \rho < \rho_{water}$ with ρ_{oil} and ρ_{water} representing the densities of oil and water, respectively. The oil and water are immiscible. If the above ball is in equilibrium in a mixture of this oil and water, which of the following pictures represents its equilibrium position?

(1) 

(2) 

(3) 

(4) 

8. Two wires are made of the same material and have the same volume. However wire 1 has crossectional area A and wire-2 has cross-sectional area $3A$. If the length of wire 1 increases by Δx on applying force F, how much force is needed to stretch wire 2 by the same amount?

(1) F (2) $4F$

(3) $6F$ (4) $9F$

9. A calorimeter of water equivalent 20 g is filled with 230 g water at 20°C. An unknown solid of mass 150 g heated to a temperature of 75°C is put into the calorimeter and the system is thermally insulated from the surroundings. The final temperature of the system is found to be 25°C. The water equivalent of the solid is

(1) 150 g (2) 75 g

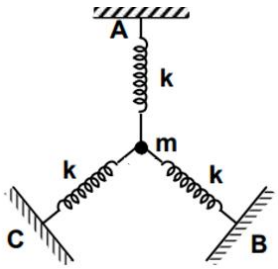
(3) 45 g (4) 25 g

10. One kg of a diatomic gas is at a pressure of 8×10^4 N/m^2. The density of the gas is 4 kg/m^3. What is the energy of the gas due to its thermal motion?

(1) 3×10^4 J (2) 5×10^4 J

(3) 6×10^4 J (4) 7×10^4 J

11. A particle of mass m is attached to three identical springs of spring constant k connected to the three vertices of an equilateral triangle ABC. The mass m is pressed against the spring connected at point C and released. The time period of small oscillations is



(1) $2\pi\sqrt{\frac{3m}{k}}$ (2) $2\pi\sqrt{\frac{m}{3k}}$

(3) $\frac{2\pi}{3}\sqrt{\frac{m}{k}}$ (4) $2\pi\sqrt{\frac{2m}{3k}}$

12. In a standing wave on a string, the spacing between nodes is Δx. If the tension in the string is doubled but the frequency of the standing waves is fixed, then the spacing between the nodes will change to

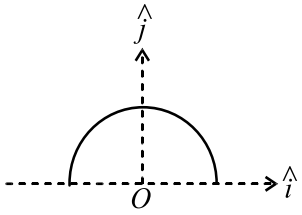
(1) $\sqrt{2}\Delta x$

(2) Δx

(3) $\Delta x/2$

(4) $\Delta x / \sqrt{2}$

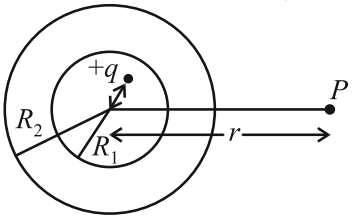
13. A thin semi-circular ring of radius r has a positive charge q distributed uniformly over it. The net field $\vec{E}$ at the centre O is



(1) $\frac{q}{4\pi^2\varepsilon_0 r^2}\hat{j}$ (2) $-\frac{q}{4\pi^2\varepsilon_0 r^2}\hat{j}$

(3) $-\frac{q}{2\pi^2\varepsilon_0 r^2}\hat{j}$ (4) $\frac{q}{2\pi^2\varepsilon_0 r^2}\hat{j}$

14. A conducting shell having inner radius R_1 and outer radius R_2 contains a charge $+q$ which is placed at a distance x from its centre. Electric field at an exterior point P which is situated at a distance r from centre of shell ($r > R_2$)



(1) Depends on x

(2) Depends on R_1

(3) Depends only or r

(4) Depends on R_2

15. A resistor 'R' and 2μF capacitor in series is connected through a switch to 200 V direct supply. Across the capacitor there is an ideal voltmeter connected. Calculate the value of R to make the reading of voltmeter to 120V in 5s after the switch has been closed. ($\log_{10}2.5 = 0.4$)

(1) $1.7 \times 10^5\ \Omega$ (2) $2.7 \times 10^6\ \Omega$

(3) $3.3 \times 10^7\ \Omega$ (4) $1.3 \times 10^4\ \Omega$

16. A current I flows in an infinitely long wire with cross section in the form of a semicircular ring of radius R. The magnitude of the magnetic induction along its axis is

(1) $\frac{\mu_0 I}{2\pi^2 R}$ (2) $\frac{\mu_0 I}{2\pi R}$

(3) $\frac{\mu_0 I}{4\pi^2 R}$ (4) $\frac{\mu_0 I}{\pi^2 R}$

17. A constant and uniform magnetic field of strength, B = 1T is established in a cylindrical region of radius 10 cm. A square conducting wire loop of size 30 cm × 30 cm is taken and moved so that all the magnetic field lines pass through this loop once. The resistance of the loop is 200 Ω. The net charge flowing through the loop is

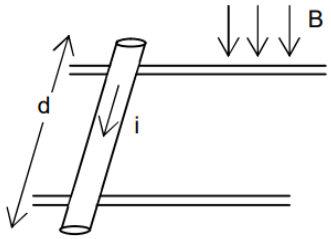
(1) 4.5×10^{-6} C (2) 4.5×10^{-4} C

(3) 3.57×10^{-4} C (4) 1.57×10^{-4}C

18. A student constructs a series RLC circuit. While operating the circuit at a frequency f he uses an AC voltmeter and measures the potential difference across each device as $(\Delta V_R)_{max}$ = 8.8 V, $(\Delta V_L)_{max}$ = 2.6 V, and $(\Delta V_C)_{max}$ = 7.4 V. Inductor in the circuit is connected next to capacitor. What result student will obtain for a measurement of combined potential difference $(\Delta V_L + \Delta V_C)_{max}$ across the inductor and capacitor.

(1) 10.0 V
(2) 7.8 V
(3) 7.4 V
(4) 4.8 V

19. A solid cylindrical uniform rod of mass 0.72 kg, length 50 cm and radius 6 cm rests on two parallel rails, as shown. The rod caries a current I = 48 A (In the direction shown) and rolls along the rails without slipping. If it starts from rest due to applications of uniform magnetic field of magnitudes 0.25 T (directed perpendicular to the rod and the rail), then the friction force between rod and rails is



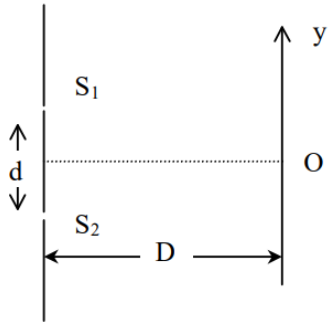
(1) 2 N
(2) 3 N
(3) 4 N
(4) 5 N

20. In an optics experiment, with the position of the object fixed, a student varies the position of a convex lens and for each position, the screen is adjusted to get a clear image of the object. A graph between the object distance (magnitude) u and the image distance (magnitude) v, from the lens, is plotted using the same scale for the two axes. A straight line passing through the origin and making an angle of 45° with the x-axis meets the experimental curve at P. The coordinates of P will be

(1) $(2f, 2f)$ (2) $\left(\frac{f}{2}, \frac{f}{2}\right)$
(3) (f, f) (4) $(4f, 4f)$

Integer Type Questions

21. Consider a usual set up of Young's double slit experiment with slits of equal intensity as shown in the figure. Take 'O' as origin and the Y axis as indicated. If average intensity between $y_1 = -\frac{\lambda D}{4d}$ and $y_2 = +\frac{\lambda D}{4d}$ equals 'n' times the intensity of maxima, then find the value of $(2n\text{-}1)\pi$

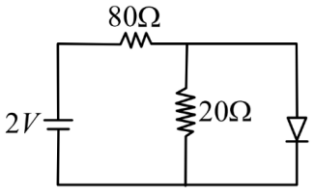


22. The orbital radius of a hydrogen atom is 0.847 nm. The electron makes a transition from this state to a lower state. How many possible wavelengths are emitted?

23. A machine gun fires a bullet of mass 40 g with a velocity 1200 ms^{-1}. The man holding it can exert a maximum force of 144 N on the gun. How many bullets can he fire per second at the most?

24. When a centimeter thick surface is illuminated with light of wavelength λ, the stopping potential is V. When the same surface is illuminated by light of wavelength 2λ, the stopping potential is $V/3$. Find the ratio of $\frac{\text{Threshold wavelength}}{\lambda}$.

25. In the circuit shown, the current through the ideal diode is (in mA)



SECTION-II (CHEMISTRY)

Single Correct Type Questions

26. 50 mL of a solution of Na_2CO_3 neutralizes 49.35 mL of 4 N HCl. The reaction is represented as

$$CO_3^{2-} + 2H^+ \rightarrow CO_2 + H_2O$$

The density of this Na_2CO_3 solution is 1.25 g mL^{-1}. The percentage of Na_2CO_3 in it is:

(1) 47.7 (2) 37.7
(3) 26.7 (4) 16.7

27. The ratio of time periods taken by electron in 1st and 3rd orbits of He^+ ion, for each revolution is ……

(1) 1 : 9 (2) 1 : 27
(3) 8 : 27 (4) 8 : 9

28. The radius of the Cl^- ion is 38% larger than that of the F^- ion but the radius of the Br^- ion is only 6.5% larger than that of the Cl^- ion. The relatively small difference in size between Cl^- and Br^- ions is due to the fact that

(1) The Br^- ion contains ten 3d electrons, which fail to shield the nuclear charge effectively.
(2) The Br^- ion contains ten 3d electrons, which shield the nuclear charge effectively.
(3) The Br^- ion contains six 4p electrons which shield the nuclear charge effectively.
(4) The Br^- ion contains ten 3d electrons and six 3p electrons, together they shield the nuclear charge effectively.

29. Which of the following is having highest bond length?

(1) NO^- (2) NO^+
(3) CN^- (4) O_2^+

30. The brown ring complex $[Fe(H_2O)_5NO]SO_4$ has oxidation number of Fe

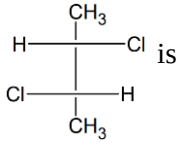
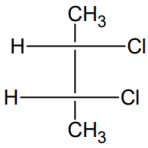
(1) +1
(2) +2
(3) +3
(4) +4

31. Which of the following statement is correct?

(1) Free energy (G) is a path function
(2) In the expression G = H – TS, TS is the part of the system's energy that is disordered, already.
(3) ΔG for a system is equal to-$T\Delta S_{Total}$ in magnitude
(4) Statement 1 is incorrect but statement 2 and 3 are correct

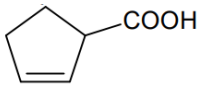
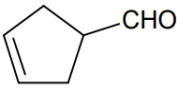
32. HCl gas is passed through an aqueous solution of 0.1 M 1 – aminopropane ($PrNH_2$) till the pH reaches 9.71. Calculate the ratio of $[PrNH_2]/[PrNH_3]^+$ in this solution:
$[K_a(PrNH_3^+) = 1.96 \times 10^{-11}]$

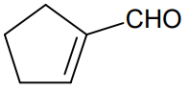
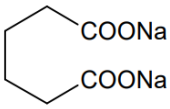
(1) 0.01 (2) 0.002
(3) 0.1 (4) 10^{-3}

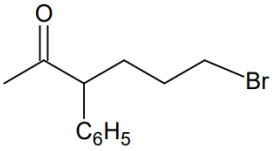
33. If the optical rotation produced by  is +36°, then the optical rotation produce by  will be

(1) – 36° (2) 0°
(3) +36° (4) Unpredictable

34. Cyclohexene on ozonolysis followed by reaction with zinc dust and water gives compound B. Compound B on further treatment with aqueous NaOH followed by heating yields compound C. The compound 'C' is:

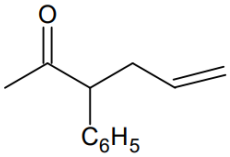
(1)  (2) 

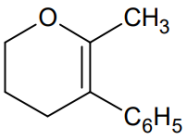
(3)  (4) 

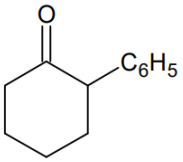
35. 

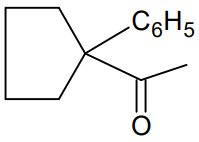
$\xrightarrow[DMSO]{CH_3CH_2ONa}$ Major product

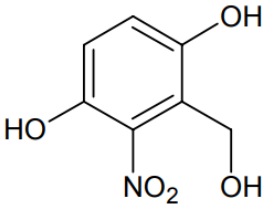
The major product is

(1) 

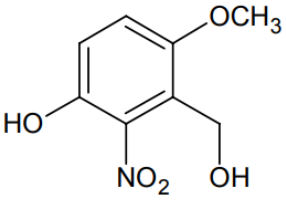
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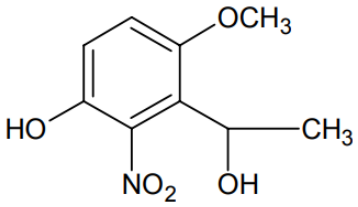
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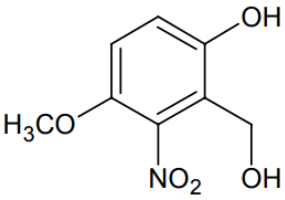
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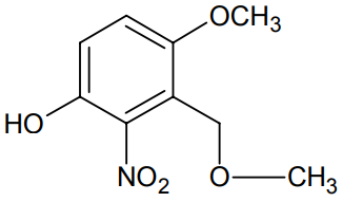
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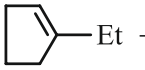
$\xrightarrow[(1\text{ eq.})]{CH_2N_2}$ A, A is :

(1) 

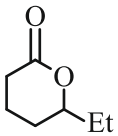
(2) 

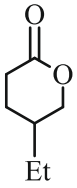
(3) 

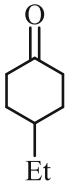
(4) 

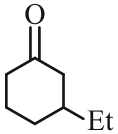
37. 

$\xrightarrow[(2)\ Ag_2O/NH_4OH\ (3)NaBH_4\ (4)H^+]{(1)\ O_3/Zn/H_2O}$ A , Product (A) is:

(1) 

(2) 

(3) 

(4) 

38. A compound (X) has the molecular formula C_3H_7NO gives Y on reaction with KOH and Br_2. Compound Y responds to mustard oil reaction. Y upon treatment with HNO_2 evolves N_2 and gives an alcohol Z which gives iodoform test. The compound X can be

(1) $C_2H_5CONH_2$ (2) CH_3CONH_2
(3) CH_3COONH_4 (4) C_2H_5NO

39. The configuration of the C-2 epimer of D-glucose is

(1) 2R, 3S, 4R, 5S (2) 2R, 3S, 4R, 5R
(3) 2S, 3S, 4R, 5R (4) 2S, 3R, 4R, 5S

40. Which metal liberates H_2 with very dilute nitric acid:

(1) Zn (2) Cu
(3) Mn (4) Fe

41. $K_2Cr_2O_7$ when reacts with cold conc. H_2SO_4 gives red crystal of

(1) CrO_4^{-2}
(2) CrO_3
(3) $Cr_2(SO_4)_3$
(4) Cr_2O_3

42. Match the following

Column- I		Column-II	
A	$[Ma_3b_2c]$	P	No. of geometrical isomer = 5
B	$[Ma_3b_3]$	Q	No. of geometrical isomers =2
C	$[Ma_3bcd]$	R	No. of geometrical isomers =4
D	$[Ma_4bc]$	S	No. of geometrical isomers = 3

(1) A→S; B→Q; C→R; D→Q
(2) A→S B→Q; C→P; D→R
(3) A→P B→Q; C→R; D→S
(4) A→Q B→S; C→P; D→R

43. At a given temp, total vapour pressure in Torr of a mixture of volatile components A and B is given by P = 120 – 75 × B.
Hence vapour pressures of pure A and B respectively (in Torr) are

(1) 120, 75
(2) 120, 195
(3) 120, 45
(4) 75, 45

44. The chlorate ion can disproportionate in basic solution according to reaction,

$$2ClO_3^- \rightleftharpoons ClO_2^- + ClO_4^-$$

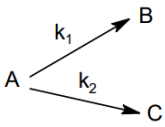
What is the equilibrium concentration of perchlorate ions from a solution initially at 0.1 M in chlorate ions at 298 K?

Given: $E^\circ_{ClO_4^-/ClO_3^-} = 0.36\,V$ and

$E^\circ_{ClO_3/ClO_2^-} = 0.33\,V$ at 298 K .

(1) 0.019 M
(2) 0.024 M
(3) 0.1 M
(4) 0.19 M

45. For first order parallel reaction k_1 and k_2 are 4 and 2 min^{-1} respectively at 300 K. If the activation energies for the formation of B and C are respectively 30,000 and 33,314 J/mol respectively. The
Temperature at which B and C will be obtained in equimolar ratio is:



(1) 757.48 K (2) 379.77 K
(3) 600 K (4) None of these

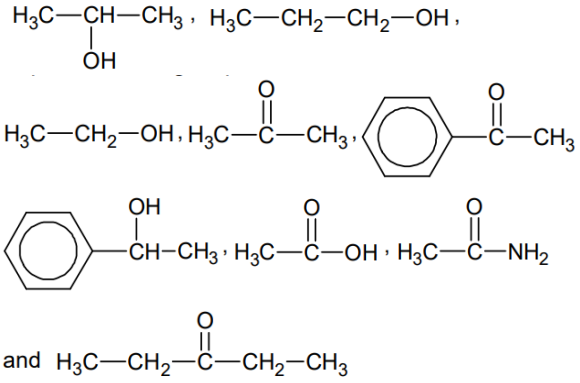
Integer Type Questions

46. How many optically active alkanes are possible with molecular formula C_7H_{16}?

47. How many of the following nitrates give positive brown ring test as well as oxygen on heating?
KNO_3, $Pb(NO_3)_2$, NH_4NO_3, $Ca(NO_3)_2$, $Sr(NO_3)_2$, $Mg(NO_3)2$, $NaNO_3$

48. A hydrocarbon $C_xH_{12}(g)$ and $O_2(g)$ were taken in a closed flask at 400 K. When sparked they reacted to form $CO_2(g)$ and $H_2O(g)$ and no reactants left after the reaction. The pressure in the flask is increased by 20%. The value of 'x' is

49. Among the following the total number of compound which give positive iodoform test is:



50. Copper reduces NO_3^- into NO and NO_2 depending upon the concentration of HNO_3 in solution. (Assuming fixed $\left[Cu^{2+}\right]$ and $P_{NO} = P_{NO_2} = 1$ bar), the HNO_3 concentration at which the thermodynamic tendency for reduction of NO_3^- into NO and NO_2 by copper is same is 10^x M. The value of x is. (Rounded-off to the nearest integer)
[Given, $E^\circ_{Cu^{2+}/Cu} = 0.34$ V, $E^\circ_{NO_3^-/NO} = 0.96$ V, $E^\circ_{NO_3^-/NO_2}$ $= 0.79$ V and at 298 K, $\frac{RT}{F}(2.303) = 0.059$]

SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. If a function $y = f(x)$ is such that $f'(x) < 0$, then the number of integral values of 'a' for which the major axis of ellipse

$$\frac{x^2}{f(a^2+2a+5)} + \frac{y^2}{f(a+11)} = 1$$

becomes x-axis is

(1) 1 (2) 2
(3) 3 (4) 4

52. b and c are the arithmetic means between a and d ($a > d > 0$) and h and k are the geometric means between a and d then

(1) bc is always greater than hk
(2) bc is always less than hk
(3) bc may be equal to hk
(4) none of these

53. If $\sin x + \sin y \geq \cos\alpha \cos x \ \forall\ x \in R$ then $\sin y + \cos\alpha$ is equal to

(1) $\frac{1}{2}$ (2) 1
(3) 2 (4) –1

54. Let $I = \int_0^1 \frac{\sin x}{\sqrt{x}}dx$ and $J = \int_0^1 \frac{\cos x}{\sqrt{x}}dx$. Then which one of the followings is true?

(1) $I > \frac{2}{3}$ and $J > 2$ (2) $I < \frac{2}{3}$ and $J < 2$
(3) $I < \frac{2}{3}$ and $J > 2$ (4) $I > \frac{2}{3}$ and $J < 2$

55. Match the following Column-I with Column-II.

	Column-I		Column -II
A	One ball is drawn from a bag containing 4 balls and is found to be white. The events that the bag contains 1 white, 2 white, 3 white and 4 white balls are equally likely. If the probability that all the balls are white is $\frac{p}{15}$ then the value of p is	P	9
B	If $x, y, z > 0$ and $x + y + z = 1$, then the least value of $\frac{5x}{2-x} + \frac{5y}{2-y} + \frac{5z}{2-z}$ is	Q	3

C	If $\sum_{K=1}^{12} 12K \cdot {}^{12}C_K \cdot {}^{11}C_{K-1}$ is equal to $\frac{12\times 21\times 19\times 17\times \ldots..\times 3}{11!}\times 2^{12}\times p$ then the value of p is	R	6

(1) A→R; B→Q; C→R
(2) A→P; B→Q; C→R
(3) A→R; B→P; C→R
(4) A→Q; B→R; C→P

56. If $a > b > c$ and the system of equations $ax + by + cz = 0$, $bx + cy + az = 0$ and $cx + ay + bz = 0$ has a non-trivial solution, then the quadratic equation $ax^2 + bx + c = 0$ has
(1) at least one positive root
(2) positive roots
(3) roots of opposite sign
(4) imaginary roots

57. Solution of the differential equation $\left(x\frac{dy}{dx}+y\right)=e^{xy-\ln x^2}\left(x\frac{dy}{dx}-y\right)$ is
(1) $\frac{y}{x}-e^{-xy}=c$ (2) $\frac{x}{y}+e^{-xy}=c$
(3) $\frac{y}{x}+e^{-xy}=c$ (4) $-\frac{x}{y}+e^{-xy}=c$

58. If $xf(x)=3(f(x))^2+2$ then
$\int\frac{2x^2-12xf(x)+f(x)}{(6f(x)-x)(x^2-(f(x))^2}dx$ is equal to
(1) $\frac{1}{(6f(x)-x)^2}+c$ (2) $\frac{1}{x^2-f(x)}+c$
(3) $\frac{1}{x^2+xf(x)}+c$ (4) none of these

59. Let $f(x)=\lim_{h\to 0}\left(\frac{(\sin(x+h))^{\ln(x+h)}-(\sin x)^{\ln x}}{h}\right)$
then $f\left(\frac{\pi}{2}\right)$ is
(1) 0 (2) 1
(3) $\ln\left(\frac{\pi}{2}\right)$ (4) none of these

60. Number of 4 digit numbers of the form $N = x_1x_2x_3x_4$ such that $2000 \le N \le 5000$, $4 \le x_2 < x_3 \le 8$ and N is multiple of 5 is
(1) 80 (2) 60
(3) 40 (4) 30

61. A biased coin is tossed repeatedly until a tail appears for the 1st time. The head is 2 times more likely to appear than tail. The probability that the number of tosses required will be more than 6, given that in 1st three tosses no tail has occurred, is
(1) $\frac{16}{81}$ (2) $\frac{32}{243}$
(3) $\frac{64}{729}$ (4) none of these

62. Let the matrices $A=\begin{bmatrix}\sqrt{3} & -2\\ 0 & 1\end{bmatrix}$ and P be any orthogonal matrix such that $Q=PAP'$ and let $R=[r_i]_{2\times 2}=P'Q^8P$ then
(1) $r_{11}=81$ (2) $r_{11}=81\sqrt{3}$
(3) $r_{11}=4\sqrt{3}$ (4) $r_{11}=-\sqrt{3}$

63. The value of $\lim_{n\to\infty}\sum_{r=1}^{n}\left(\sum_{t=0}^{r-1}\frac{1}{5^n}\cdot {}^nC_r\cdot {}^rC_t\cdot 3^t\right)$ is equal to
(1) 4 (2) 3
(3) 1 (4) none of these

64. If $f(x)=(2x-3\pi)^{25}+\frac{4}{3}x+\cos x$ and $g(x)$ is the inverse of $f(x)$, then the value of $\frac{d}{dx}(g(x))$ at $x = 2\pi$ is
(1) $\frac{7}{3}$ (2) $\frac{3}{7}$
(3) $\frac{30\pi^{25}+4}{3}$ (4) none of these

65. The 19th term of the sequence $1,\frac{1}{3},\frac{1}{5},\frac{1}{7}....$ is:
(1) $\frac{1}{35}$ (2) $\frac{1}{37}$
(3) $\frac{1}{39}$ (4) $\frac{1}{41}$

66. If z is a complex number then $|z| = 1$ represents
(1) Circle (2) Straight line
(3) Ellipse (4) Parabola

67. The point of intersection of the lines $y = x$ and $x + y = 2$ is
(1) (2, 2)
(2) (1, 2)
(3) (2, 1)
(4) (1, 1)

68. Let $x, y, z \in R$. The number of integral values that x can take if $x + y + z = 5$ and $x^2 + y^2 + z^2 = 9$, is

(1) 0
(2) 1
(3) 2
(4) infinitely many

69. The number of solutions of the equations $2\sin^2 x + \sin^2 2x = 2$, $\sin 2x + \cos 2x = \tan x$ in $[0, 4\pi]$ satisfying the condition $2\cos^2 x + \sin x \leq 2$ is

(1) 4
(2) 6
(3) 8
(4) 2

70. If $|z| = 1$ and $z \neq 0$, then the points $z + \frac{1}{z}$ lie on a

(1) Straight line not through the origin
(2) Circle
(3) Line segment lying on the x-axis
(4) Line segment lying on the y-axis

Integer Type Questions

71. Number of integers not in the domain of $\log \frac{x-1}{x-4}$ is

72. If $y = f(x)$ satisfies $f(x + 1) + f(z - 1) = f(x + z)\ \forall\ x, z \in R$ and $f(0) = 0$ and $f'(0) = 4$ then $f(2)$ is equal to _____.

73. If $-\frac{\pi}{2} < \alpha_1 < \alpha_2 < \alpha_3 < \frac{\pi}{2}$, then number of values of $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ satisfying $(\tan\theta - \tan\alpha_1)(\tan\theta - \tan\alpha_2)(\tan\theta - \tan\alpha_3) = \sum(\tan\theta - \tan\alpha_i)$ is _____.

74. A quadratic equation with integral coefficients has two prime numbers as its roots. If the sum of the coefficients of the equation is prime, then the sum of the roots is _____.

75. Let $I_1 = \int_0^1 \frac{e^x}{1+x}dx$ and $I_2 = \int_0^1 \frac{x^2}{e^{x^3}(2-x^3)}dx$.

Then $\frac{I_1}{e \cdot I_2}$ is equal to



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